Docker Swarm Complete Guide (Debian VM)

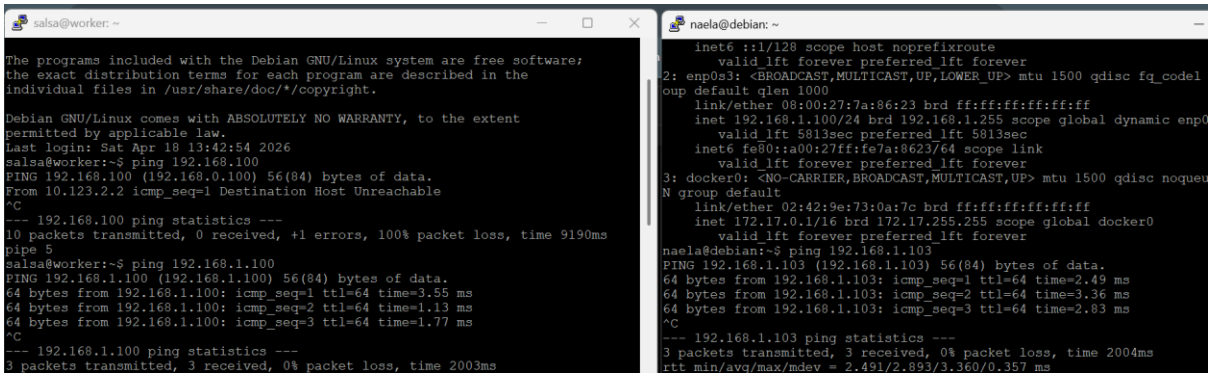
Berikut merupakan dokumentasi hasil praktikum pengerjaan Docker swarm dengan OS Debian pada virtual box.

Pra syarat sebelum praktikum:

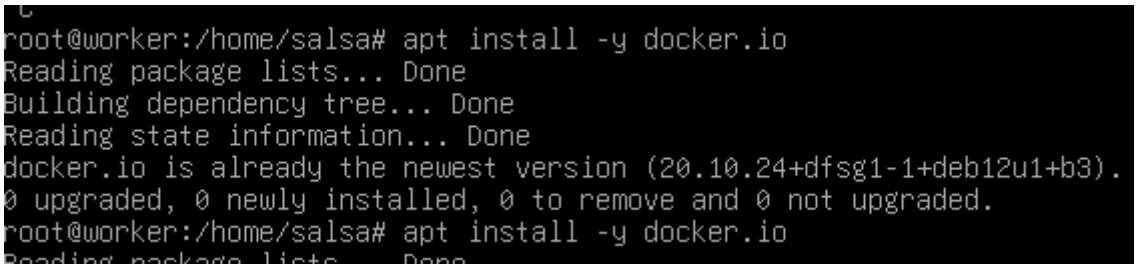
- Memiliki 2 Virtual Machine dengan OS Debian
- Jaringan Internet Stabil dan Lancar
- Memiliki Ruang Penyimpanan kosong minimal 20 GB
- Menginstall aplikasi remote server PuTTY

Instalasi dan Setup

Pastikan dapat ping antar perangkat (1 jaringan) dan TTL

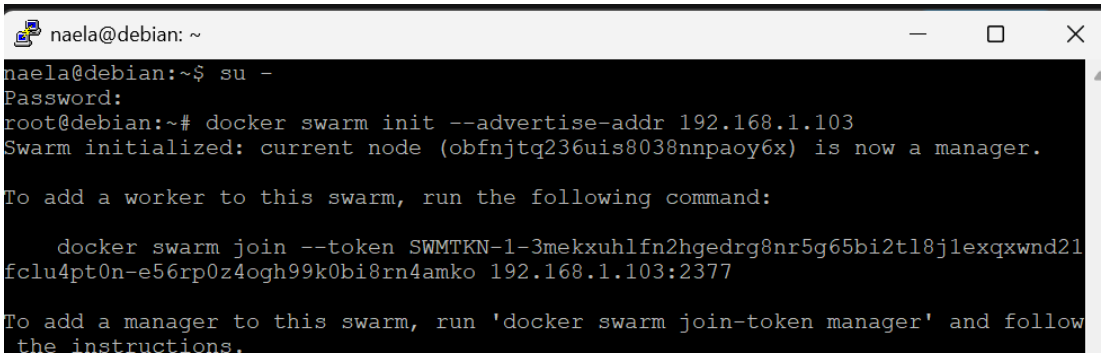


Install docker pada kedua VM



Setup Swarm

Pada VM manager buat swarm dengan memasukan IP manager



Join worker dengan masukan token pada VM workerna

```
salsa@worker: ~
root@worker:~# docker swarm join --token SWMTKN-1-52ilepg7yuhuxza02e05p4cwo2inxm
u3sgon26978g7927acqh-30sahicbx5kry3z1htiomtldm 192.168.1.100:2377
This node joined a swarm as a worker.
root@worker:~#
```

Check Cluster, pastikan sudah Ready

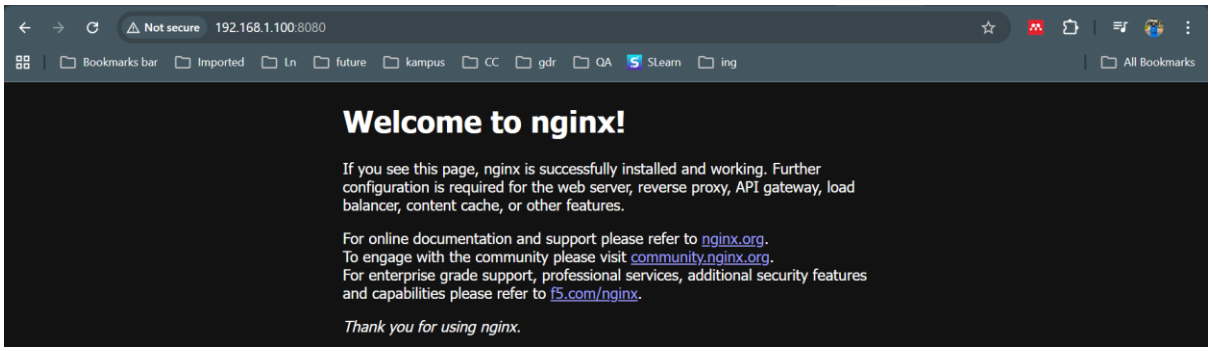
```
root@debian:~# docker node ls
ID                            HOSTNAME   STATUS    AVAILABILITY   MANAGER STATUS
   ENGINE VERSION
zq2p9iimorxjyp5v3zwqz7ogh *   debian     Ready     Active         Leader
   20.10.24+dfsg1
tu8mhwrtlol09o5hwjv6spxep     worker     Ready     Active
   20.10.24+dfsg1
```

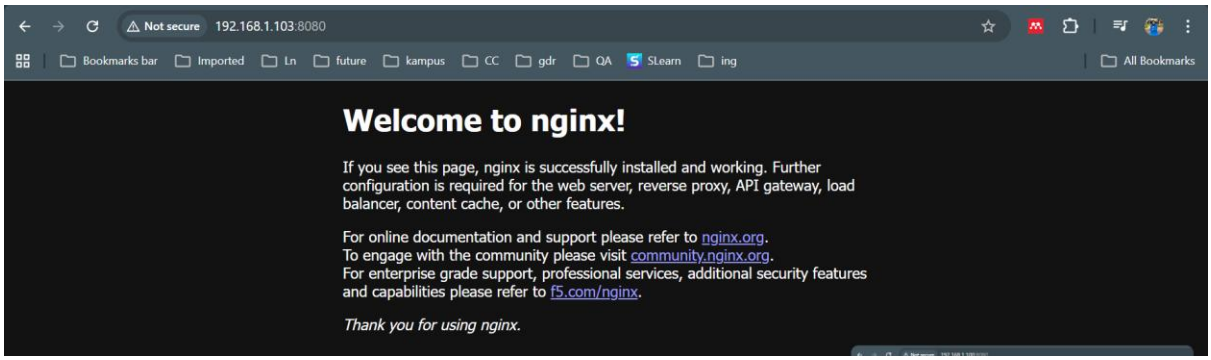
Test Service

Buat container nginx sebagai web-server, lalu pastikan service berhasil dijalankan

```
root@debian:~# docker service create --name web --replicas 3 -p 8080:80 nginx
18r10tegfrhmsyznx2quo7nbh
overall progress: 3 out of 3 tasks
1/3: running
2/3: running
3/3: running
verify: Service converged
root@debian:~# docker service ls
ID             NAME      MODE         REPLICAS   IMAGE          PORTS
18r10tegfrhm   web       replicated   3/3        nginx:latest   *:8080->80/tcp
root@debian:~#
```

Check ip manager dan worker pada browser apakah web-server berhasil dijalankan





Distribusi Container

Melihat task container yang berjalan

```
root@debian:~# docker service ps web
ID             NAME        IMAGE          NODE      DESIRED STATE   CURRENT STAT
E            ERROR     PORTS
sbuy887pu0kx   web.1       nginx:latest   debian    Running         Running 4 mi
nutes ago
dw8pyrydd1lq    \_ web.1   nginx:latest   worker    Shutdown        Shutdown 4 m
inutes ago
7eqd4aw6p3tv   web.2       nginx:latest   worker    Running         Running 4 mi
nutes ago
lt27lmsez4m7    \_ web.2   nginx:latest   debian    Shutdown        Shutdown 4 m
inutes ago
fa6guciqs1hm   web.3       nginx:latest   worker    Running         Running 4 mi
nutes ago
bbcybfj52633    \_ web.3   nginx:latest   worker    Shutdown        Shutdown 4 m
inutes ago
```

Membuat Scale menjadi 5 web-server dengan $docker service scale web=5

```
root@debian:~# docker service ps web --filter "desired-state=running"
ID             NAME      IMAGE          NODE      DESIRED STATE   CURRENT STATE           ERROR     PORTS
sbuy887pu0kx   web.1     nginx:latest   debian    Running         Running 12 minutes ago
7eqd4aw6p3tv   web.2     nginx:latest   worker    Running         Running 12 minutes ago
fa6guciqs1hm   web.3     nginx:latest   worker    Running         Running 12 minutes ago
dvdes4lxf382   web.4     nginx:latest   debian    Running         Running 2 minutes ago
8nn0ue7fwlf9   web.5     nginx:latest   debian    Running         Running 2 minutes ago
```

```
root@debian:~# docker service ps web
ID             NAME        IMAGE          NODE      DESIRED STATE   CURRENT STAT
E            ERROR     PORTS
sbuy887pu0kx   web.1       nginx:latest   debian    Running         Running 11 m
inutes ago
dw8pyrydd1lq    \_ web.1   nginx:latest   worker    Shutdown        Shutdown 11
minutes ago
7eqd4aw6p3tv   web.2       nginx:latest   worker    Running         Running 10 m
inutes ago
lt27lmsez4m7    \_ web.2   nginx:latest   debian    Shutdown        Shutdown 11
minutes ago
fa6guciqs1hm   web.3       nginx:latest   worker    Running         Running 10 m
inutes ago
bbcybfj52633    \_ web.3   nginx:latest   worker    Shutdown        Shutdown 11
minutes ago
dvdes4lxf382   web.4       nginx:latest   debian    Running         Running 39 s
econds ago
8nn0ue7fwlf9   web.5       nginx:latest   debian    Running         Running 39 s
econds ago
root@debian:~#
```